

Mastering Event Delegation in JavaScript

Understand how to handle multiple events efficiently like a pro.

Topics Covered

What you'll learn:

- What is Event Delegation?
- Why is it useful?
- How bubbling works
- Real-world DOM example (vanilla JS)
- Common mistakes



Swipe to explore



What is Event Delegation?

Event Delegation is a technique where you attach a **single event listener to a parent**, and handle events on its children using **event bubbling**.

Instead of attaching 10 listeners to 10 buttons, use 1 listener on their parent.

Why Use Event Delegation?

- Better performance
- Cleaner code
- Dynamically added elements still work
- Prevents memory leaks in SPAs

How Event Bubbling Works?

*Events in JavaScript propagate in **three phases**:*

1. **Capturing phase** (rarely used)
2. **Target phase** (event hits the target element)
3. **Bubbling phase** (event bubbles up to ancestors)

*We use bubbling in **event delegation**.*



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Real Example 1 – Vanilla JavaScript



```
1  //html
2  <ul id="list">
3    <li>Item 1</li>
4    <li>Item 2</li>
5    <li>Item 3</li>
6  </ul>
7
8  //js
9  document.getElementById('list').addEventListener('click', function(e) {
10    if (e.target.tagName === 'LI') {
11      console.log(`You clicked on ${e.target.textContent}`);
12    }
13  });
14
15  //output
16  //You clicked on Item 1
17
```



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Real Example 2 – Vanilla JavaScript



```
1  //html
2  <div id="button-container">
3    <button data-value="Save">Save</button>
4    <button data-value="Delete">Delete</button>
5    <button data-value="Edit">Edit</button>
6  </div>
7
8  //js
9  document.getElementById('button-container').addEventListener('click', function (e) {
10    // Ensure the clicked element is a button
11    if (e.target.tagName === 'BUTTON') {
12      const value = e.target.dataset.value; // Get the data-value attribute
13      console.log(`You clicked: ${value}`);
14    }
15  });
16
17  // Output Example (in Console):
18  // Clicking the Save button:
19  // You clicked: Save
20
21  // Clicking the Delete button:
22  // You clicked: Delete
23
```

Best Practices

- Use **event.target** carefully — it may be a nested span or icon.
- Add checks (like **tagName**, **className**, **dataset attributes**).
- Avoid deeply nested delegation unless necessary.

 **Thanks for Reading!**

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